AI.401

Evolutionary Robotic

COURSE OVERVIEW

Evolutionary algorithms for robot design and behavior: morphology, control co-optimization, and artificial life.



ILLUSTRATIVE HERO IMAGERY

EVOLUTIONARY ROBOTICS

1

SESSION

60 MINUTES

ORIENTATION: EVOLVING ROBOTS

KEY TOPICS

- Evolutionary computation
- Evolutionary robotics
- Morphology and control co-design
- Artificial life

TEACHING FOCUS

Let robots design themselves.

KEY TAKEAWAY Evolution discovers designs humans miss.

EVOLUTIONARY ROBOTICS

2

SESSION

60 MINUTES

GENETIC ALGORITHMS

KEY TOPICS

- Encoding and genomes
- Selection, crossover, mutation
- Fitness functions
- GA for robotics

TEACHING FOCUS

Search design spaces with evolution.

KEY TAKEAWAY Genetic algorithms optimize by breeding solutions.

EVOLUTIONARY ROBOTICS

3

SESSION

60 MINUTES

EVOLVING MORPHOLOGY

KEY TOPICS

- Evolving robot bodies
- Sim-to-real bodies
- Modular and soft robots
- Embodied evolution

TEACHING FOCUS

Design bodies and brains together.

KEY TAKEAWAY Body and mind evolve as one.

EVOLUTIONARY ROBOTICS

4

SESSION

60 MINUTES

EVOLVING CONTROL

KEY TOPICS

- Evolving neural controllers
- Behavioral diversity
- Novelty search
- Open-endedness

TEACHING FOCUS

Evolve robot behavior directly.

KEY TAKEAWAY Evolution can invent unexpected behaviors.

EVOLUTIONARY ROBOTICS

5

SESSION

60 MINUTES

EVOLUTIONARY MULTI-OBJECTIVE

KEY TOPICS

- Pareto optimization
- Trade-offs in design
- NSGA-II
- Conflicting objectives

TEACHING FOCUS

Balance competing design goals.

KEY TAKEAWAY Good design balances trade-offs.

EVOLUTIONARY ROBOTICS

6

SESSION

60 MINUTES

EVOLUTIONARY ROBOTICS & REVIEW

KEY TOPICS

- Evolution demo
- Course recap and assessment
- Toward robot ethics (AI.402)
- Evolved robot showcase

TEACHING FOCUS

Evolve a robot in simulation.

KEY TAKEAWAY Evolution is creativity with selection.